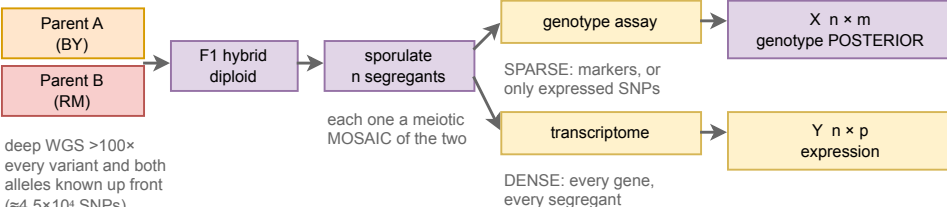
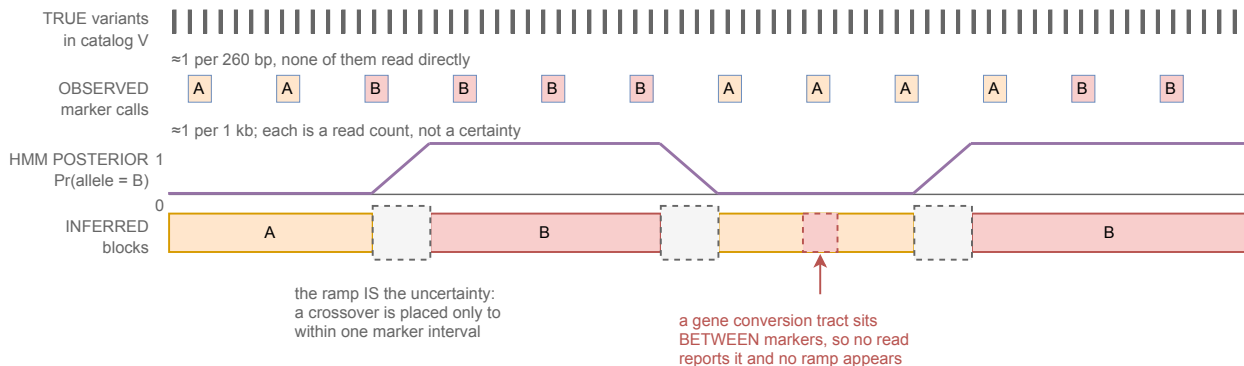


a The cross, and the two things measured on every segregant



The asymmetry. The phenotype is read directly. The genotype never is: it is INFERRED from sparse observations as a mosaic of two known genomes. Panel b is that inference; panels d and e are what it costs.

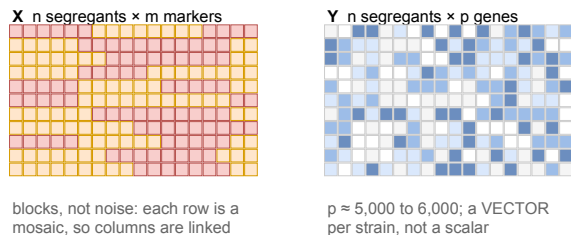
b One segregant, one chromosome: from sparse reads to a posterior



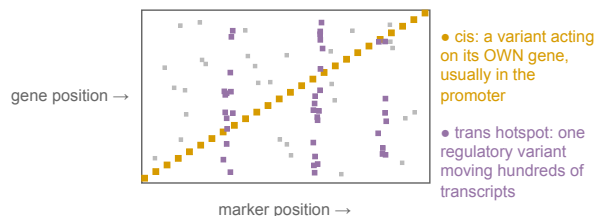
Why sparse reads still suffice. The segregant is never sequenced de novo. At each locus it is ASSIGNED to one of two already-sequenced genomes. Meiosis makes ≈ 90 crossovers, so a genome is $\approx 10^2$ blocks, and one marker fixes every cataloged variant in its block. The task is locating $\approx 10^2$ boundaries, not determining $\approx 10^4$ values.

What no marker density fixes. Boundary intervals leave $\approx 0.8\%$ of cataloged sites ambiguous. Non-crossover gene conversion (≈ 46 per meiosis, ≈ 2 kb) affects more bases than that and is invisible to sparse markers. De novo mutation, aneuploidy and non-reference sequence sit outside the mosaic model entirely.

c What the two matrices look like



d The resulting map: cis on the diagonal, trans in bands



e What that means for a record

Stored. The pair (genotype, environment) $\rightarrow$ phenotype, one record per segregant per condition. The phenotype side needs nothing new: RNASeqExpressionPhenotype and PseudobulkExpressionPhenotype already exist. The genotype side should hold the MOSAIC, $\approx 10^1$ - 10^2 intervals of (chr, start, end, parent, p) against two hash-pinned parent assemblies, rather than $\approx 10^4$ variants. That stores the posterior as a posterior, is lossless, and keeps linkage explicit.

Not stored, and one field with nowhere to go. The QTL table is an estimate conditional on method, threshold and marker density, with no source value to check, so it is not a Phenotype. And GenePerturbation REQUIRES systematic_gene_name: a cis-eQTL is usually an intergenic promoter variant with no gene to attach to. The same gap blocks every reporter assay, so the two are one decision.